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# =====
# AIM:
# Image Acquisition Noise Removal using Median Filter
# Google Colab
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

print("Upload an image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read Image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Step 1 : Simulate Sensor Noise (Salt & Pepper)
# -----

noise_img = img.copy()

prob = 0.03

random = np.random.rand(*img.shape[:2])

noise_img[random < prob] = 0          # Black pixels
noise_img[random > 1-prob] = 255      # White pixels

# -----
# Step 2 : Remove Noise
# -----

filtered = cv2.medianBlur(noise_img,5)

# -----
# Display Images
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(noise_img)
plt.title("Sensor Noisy Image")
plt.axis("off")

plt.subplot(1,3,3)

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plt.imshow(filtered)
plt.title("Noise Removed")
plt.axis("off")

plt.show()

# -----
# Output
# -----

print("\n===== IMAGE ACQUISITION REPORT =====")
print("Input Image      : Successfully Loaded")
print("Sensor Noise Added  : Salt & Pepper Noise")
print("Noise Removal       : Median Filter")
print("Image Quality        : Improved")
print("Application         : Camera Sensor Processing")
print("Result               : Noise Reduced Successfully")
print("=====")

```

Upload an image

q8.jpg

q8.jpg(image/jpeg) - 121504 bytes, last modified: 7/15/2026 - 100% done

Saving q8.jpg to q8.jpg

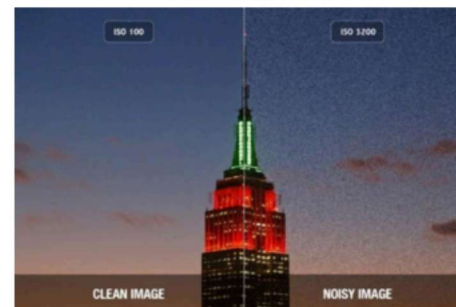
Original Image



Sensor Noisy Image



Noise Removed



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===== IMAGE ACQUISITION REPORT =====
Input Image      : Successfully Loaded
Sensor Noise Added  : Salt & Pepper Noise
Noise Removal     : Median Filter
Image Quality      : Improved
Application       : Camera Sensor Processing
Result            : Noise Reduced Successfully
=====

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